

3D Volume Image Analysis for Biologists

Segment Anything as an assistant for quick drawing of ground truth

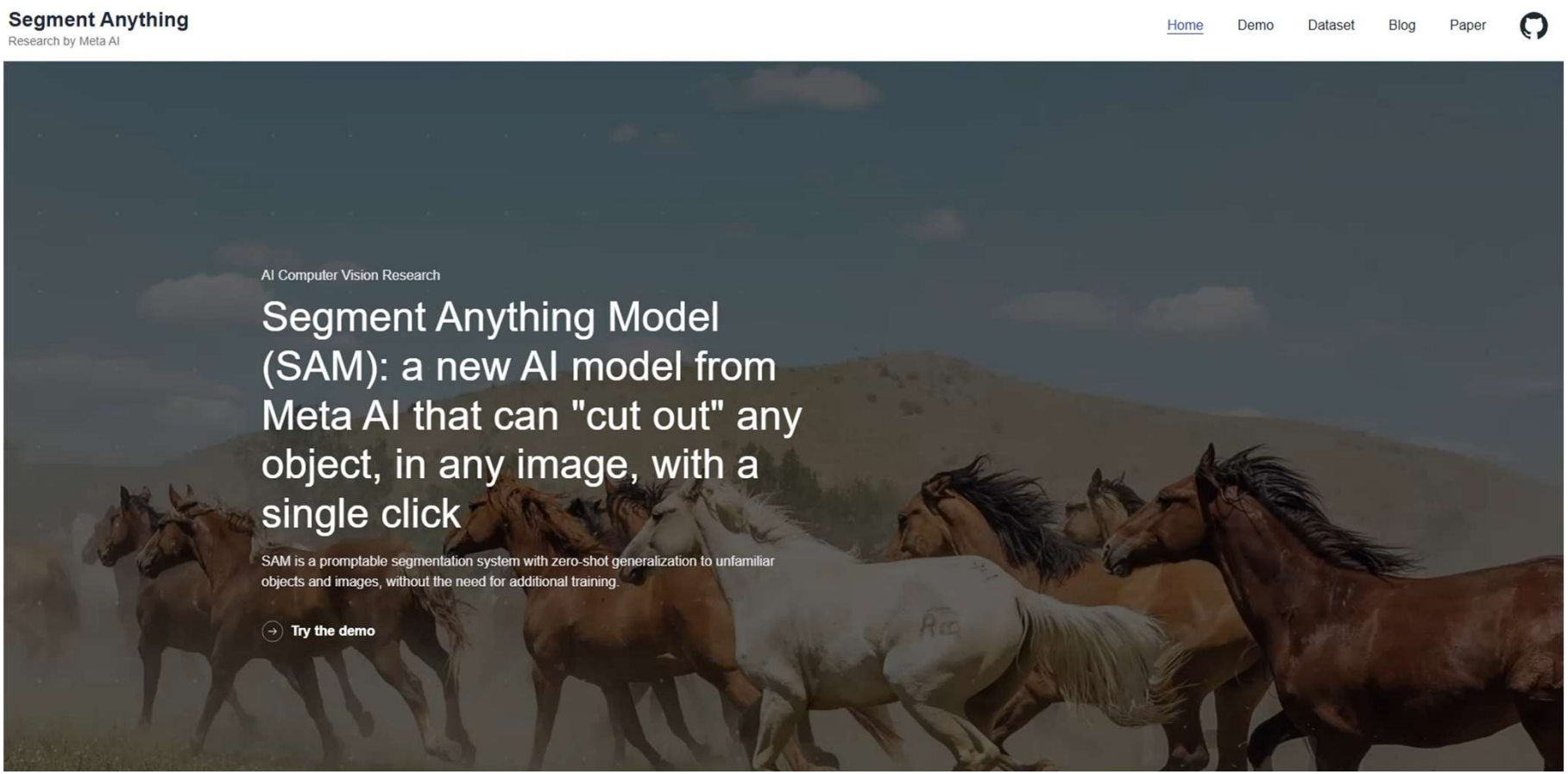
NUS National University of Singapore
Microscopy Cluster
Yong Loo Lin School of Medicine



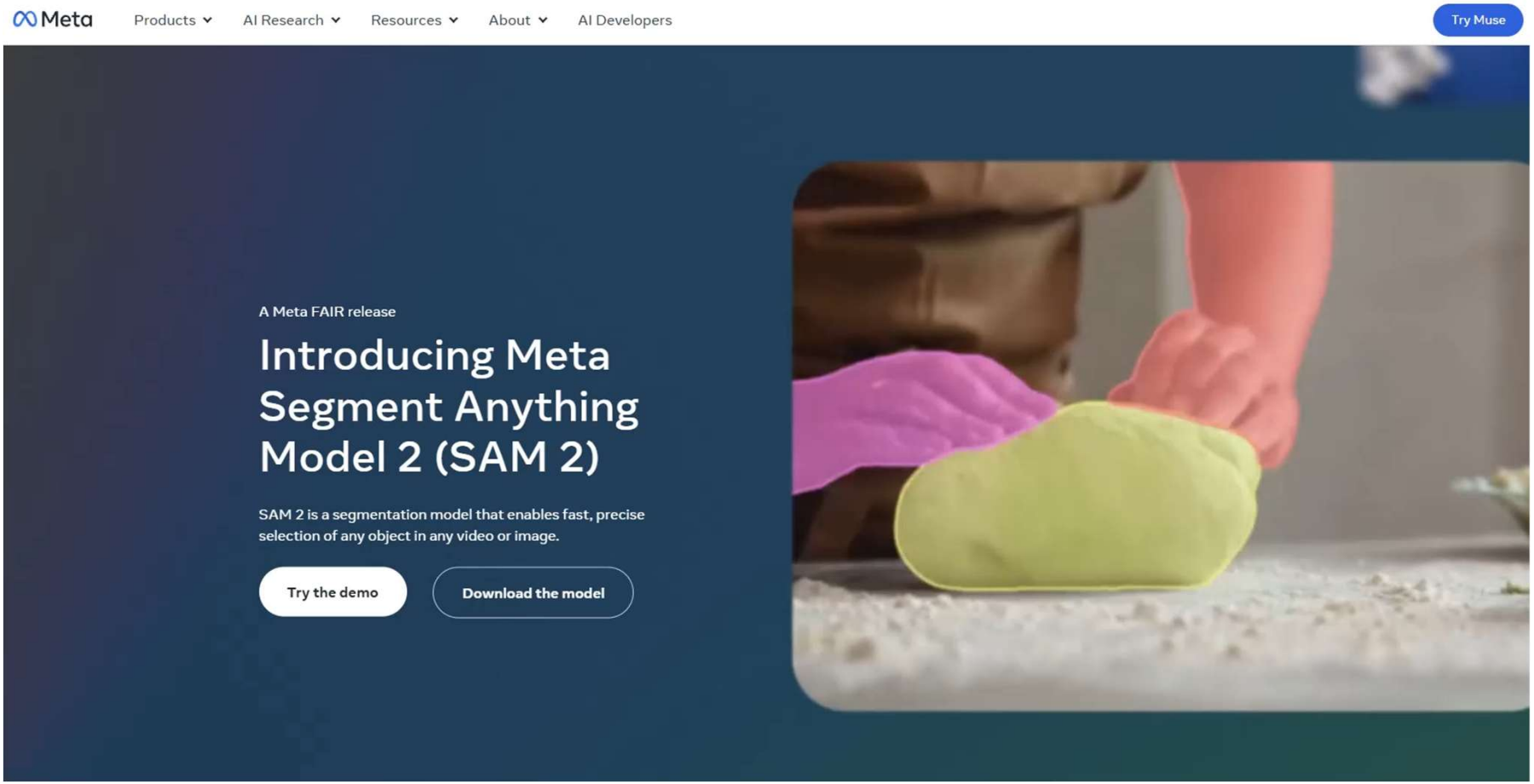
Content

- SAMJ
- Napari-SAM
- Work on your own stuff

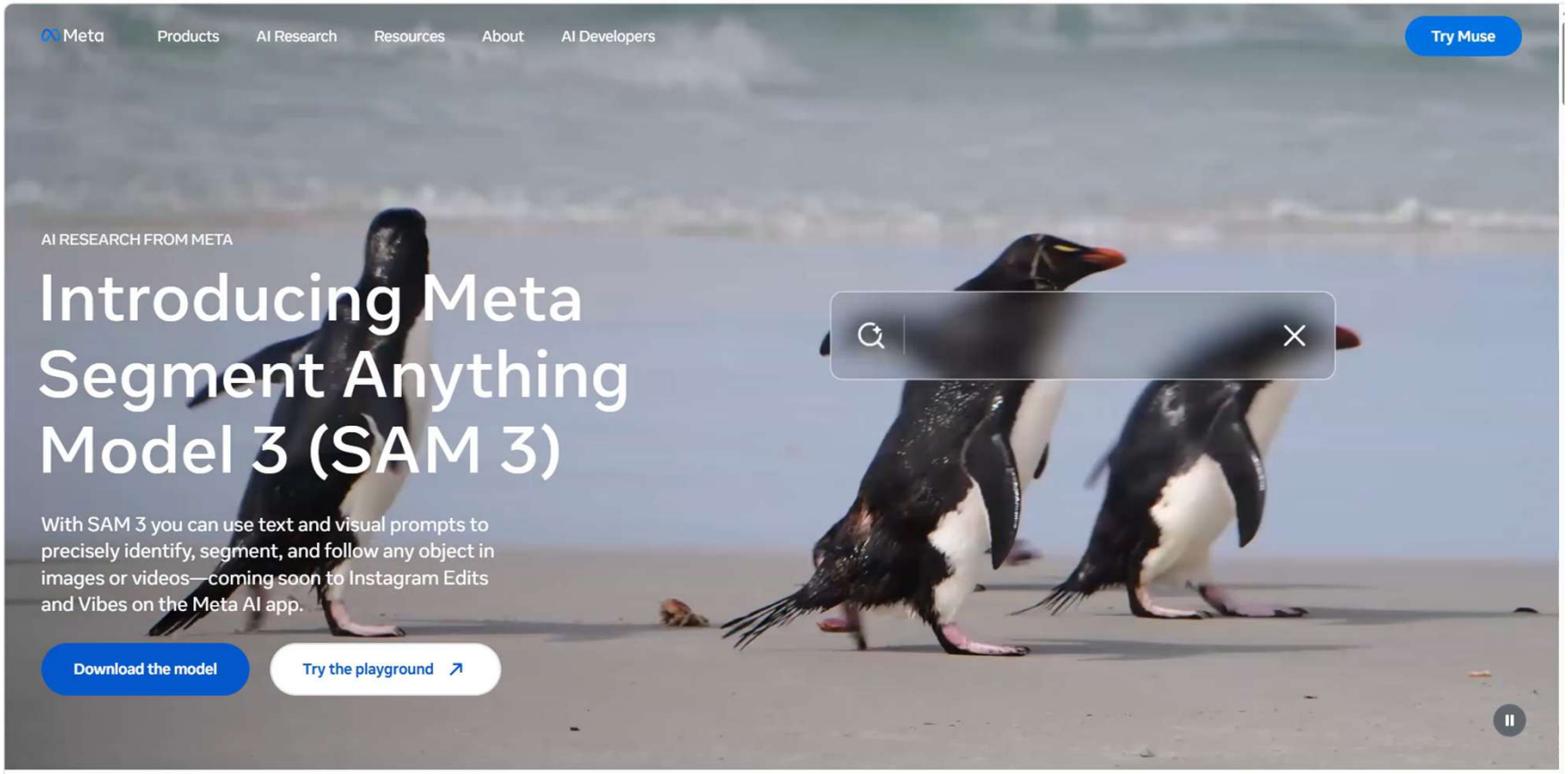
Segment Anything v1 by Meta (Facebook)



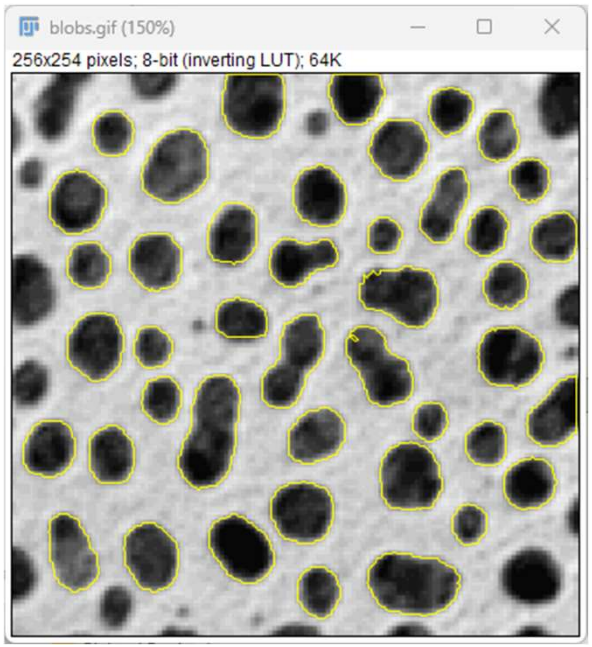
Segment Anything v2 by Meta (Facebook)

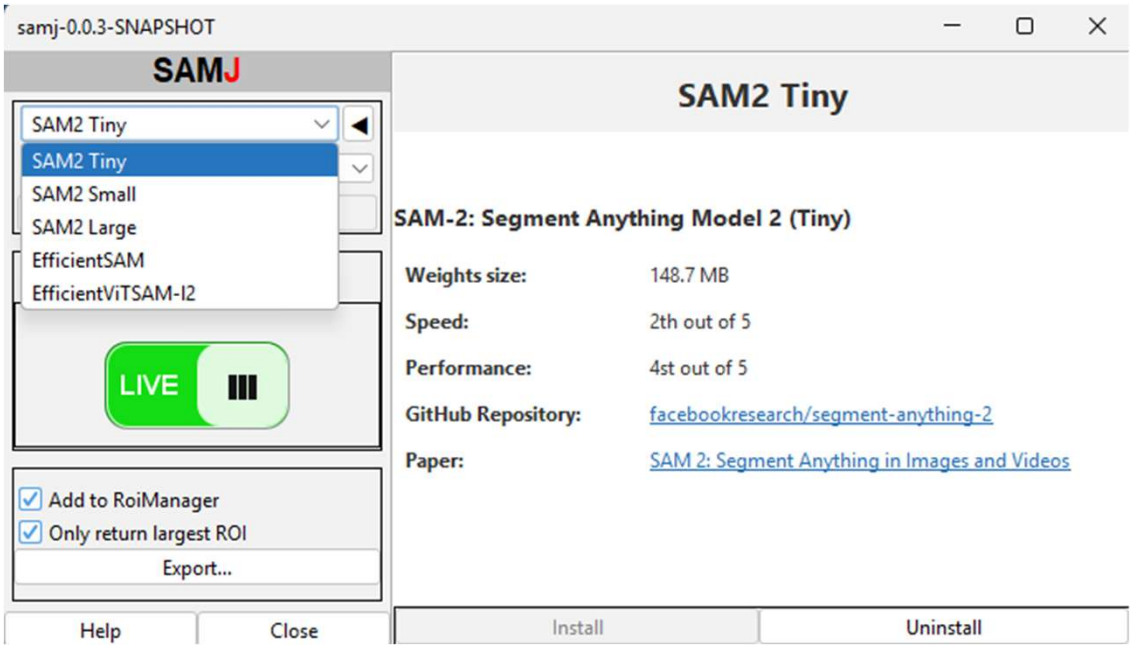


Segment Anything v3 by Meta (Facebook)



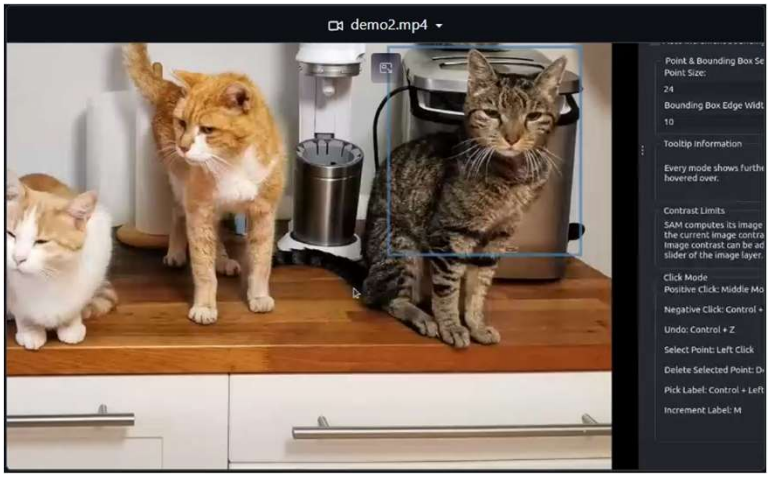
SAMJ – SAMv2 (limited to v1 use)



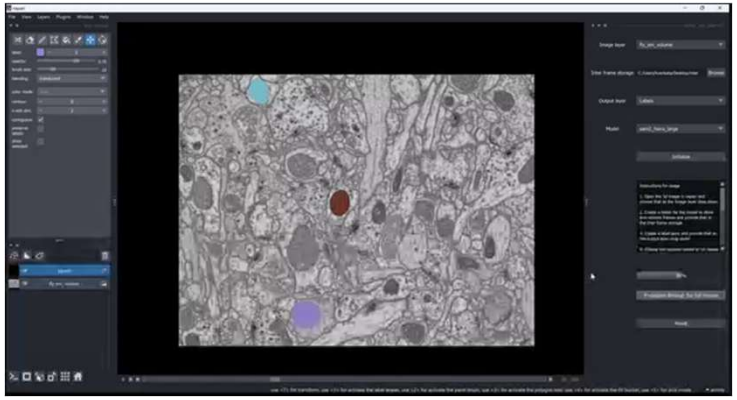


Napari-SAM

- Napari-sam
  - https://github.com/MIC-DKFZ/napari-sam



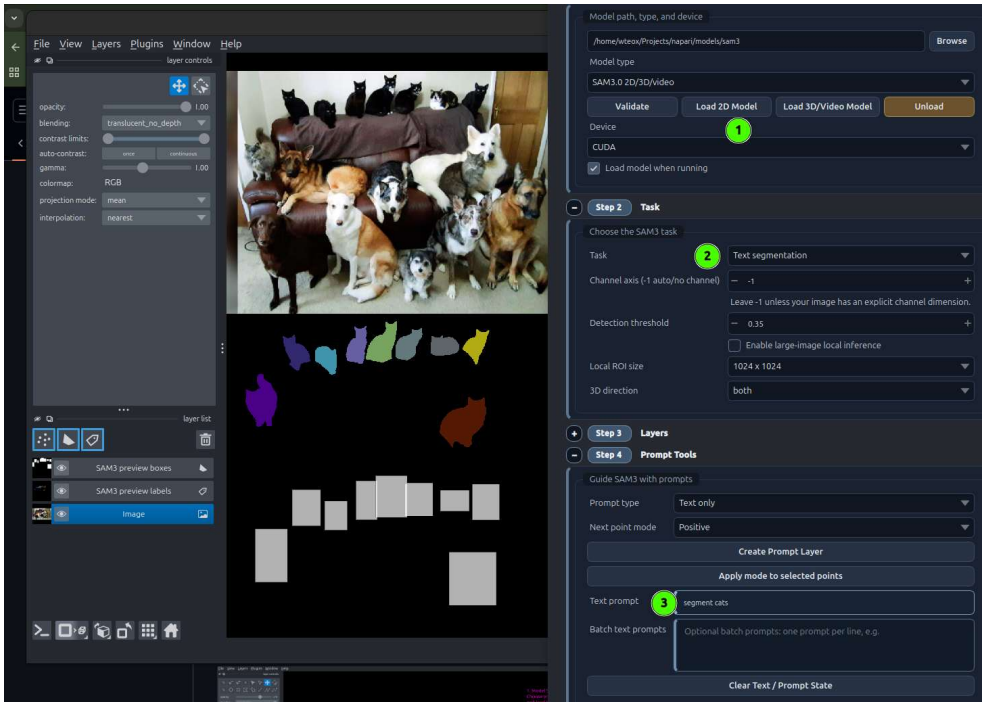
- Napari-SAMV2 (requires git)
  - https://github.com/Krishvraman/napari-SAMV2

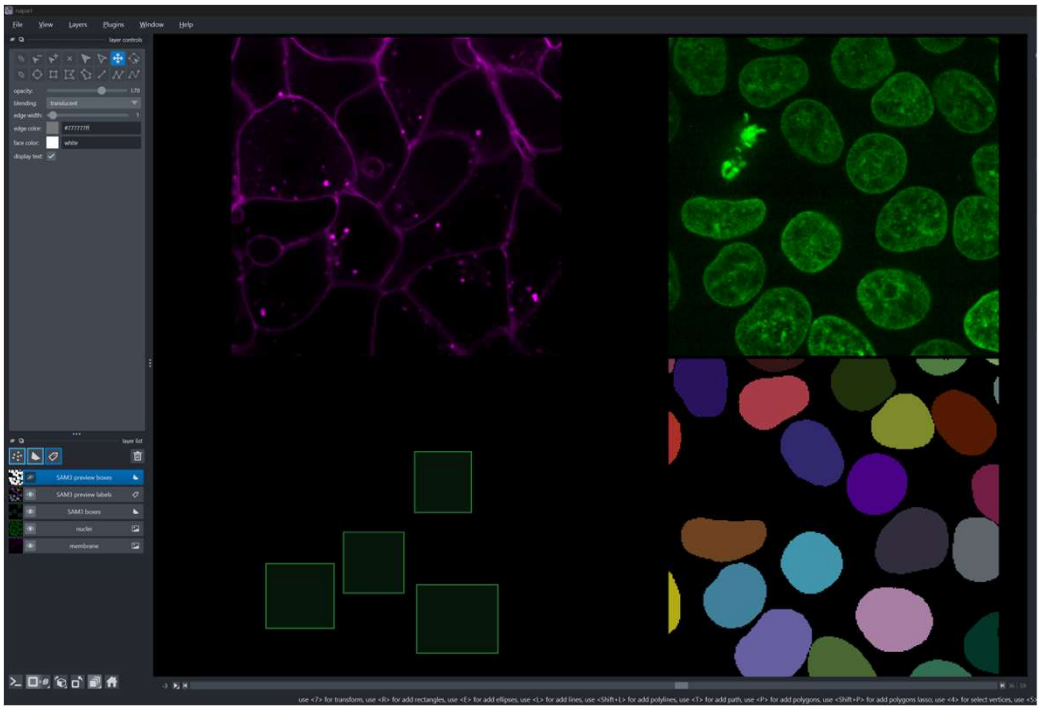




Napari-SAM

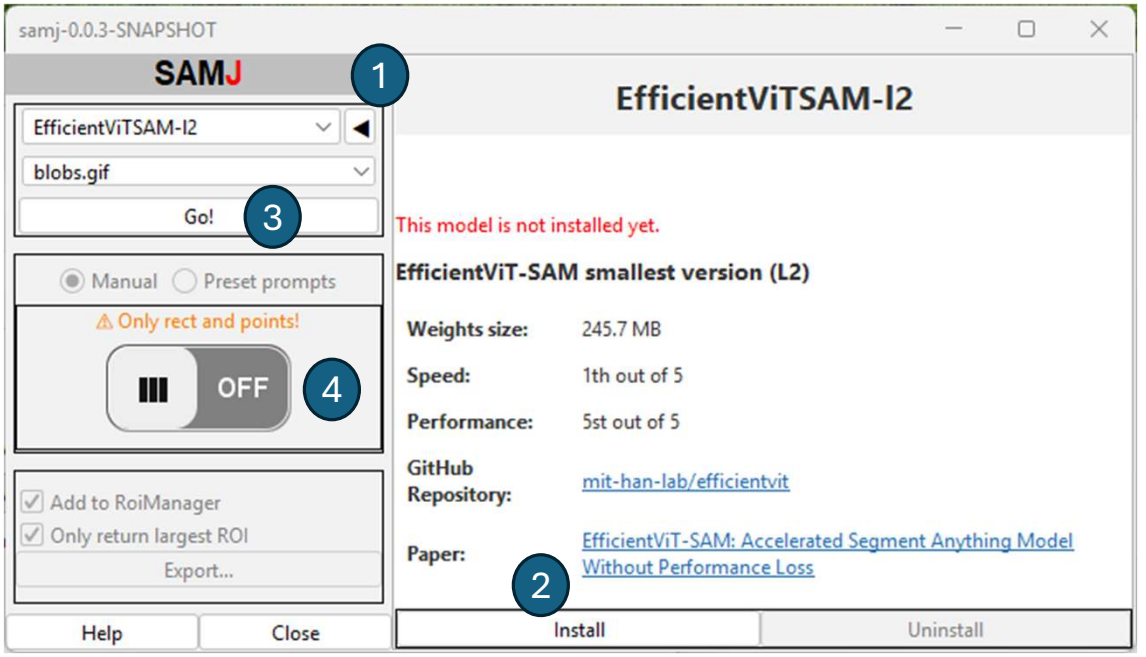
- Napari-sam3-assistant (GPU only)
  - https://github.com/wulinteousa2-hash/napari-sam3-assistant





SAMJ – SAMv2 (limited to v1 use)

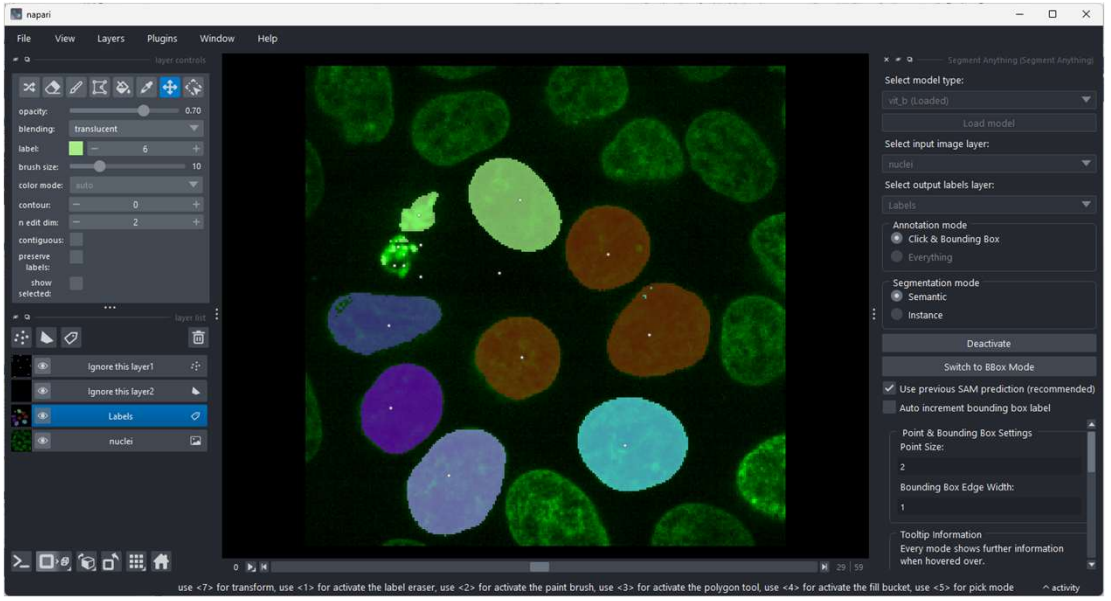
Open one of the Practice image
Select EfficientViTSAM-I2
Click on the arrow and Install (few minutes)
Click on Go!
Toggle Live



Napari-SAM

- Napari-sam (v1)
  - https://github.com/MIC-DKFZ/napari-sam

- Open Anaconda Prompt
  conda create --name env-napari-sam python=3.9
  conda activate env-napari-sam
  pip install napari[pyqt5]==0.5.6
  pip install segment-anything-py
  pip install napari-sam



- Start napari
  - Start the Segment Anything plugin
  - Open Stack_patches.tif
  - Select vit-b model then Activate
  - Create and select a label layer
  - Click on each individual mitochondria
    - Press = to increase label id (if click-mode)
    - Use the wheel-click to include a mitochondria (Ctrl + wheel-click to exclude)
    - Switch to Bounding Boxes if needed

Thanks for attending

https://ppms.asia/medicine-nus/start/
3 powerful workstations



Windows 11
16-cores CPU
128GB of RAM
Nvidia RTX 5090
4TB NVME



Windows 11
16-cores CPU
128GB of RAM
Nvidia RTX A4000
1TB NVME



Windows 11
64-cores CPU
768 GB of RAM
3x Nvidia RTX 5090
4TB NVME



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3D VOLUME IMAGE ANALYSIS COURSE FOR BIOLOGIST

PROGRAM

DATE	TOPIC	LOCATION
Mon 12-Oct-26 (Morning)	Introduction to Anaconda Basics of Napari Basics of Fiji / ImageJ	MD1-03-01C - TAHIR BLDG
Mon 12-Oct-26 (Afternoon)	Lecture on multidimensional analysis Hands-on image processing using Fiji Hands-on Image processing using Napari	MD6-03-04 - CeTM
Wed 14-Oct-26 (Morning)	How to use and retrain EMPANADA How to use and retrain Cellpose-SAM	MD1-03-01B - TAHIR BLDG
Wed 14-Oct-26 (Afternoon)	SAM3/SAMJ, a tool for quick ground truth creation	MD1-03-01B - TAHIR BLDG

Question and information, email rdemets[at]nus.edu.sg

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Thanks for attending

Survey Volume Image Analysis for Biologists - 2026

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